

Mapping unpaid-care and Economic Activity Across England and Wales: A Multi-Scale Visual Analytics Study, 2011–2021

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Abstract—Around five million people across England and Wales provide unpaid-care worth £162 billion each year, yet between 2011 and 2021 the overall share of residents providing any care fell from 11.4% to 9.0% while those providing 20 or more hours per week rose from 1.5% to 1.9%. This project was built for public health analysts and care commissioners who need to understand where that divergence between falling prevalence and rising intensity has landed hardest. A four-dashboard Tableau workbook examines census data at two geographic scales, combining dimensionality reduction, neighbourhood clustering and Bayesian residual modelling to connect national patterns to neighbourhood-level detail.

Every one of the twelve Local Authorities with the largest rises in intensive care, including Hartlepool and Eastbourne, also saw economic inactivity increase over the same decade. Local Authority averages masked wide neighbourhood variation, with Newcastle upon Tyne spanning 3.63 percentage points across its neighbourhoods despite sitting close to the national mean of 4.6%. The highest intensive-care rate of 6.5% appeared not in the oldest retirement-heavy areas but in a health-limited cluster, suggesting that intensive care aligned more closely with long-term sickness and disability than with age alone. The most ethnically diverse areas consistently recorded the lowest care rates, which challenges common assumptions about informal family-based care. In East Lindsey, three neighbourhoods still exceeded model expectations by 2.4 to 3.1 percentage points even after accounting for their already old and inactive populations. A formative evaluation showed that usability depended on visible comparison rules rather than analytical complexity, which is a finding that generalises well beyond this particular topic.

I. INTRODUCTION

Every day, around five million people across England and Wales quietly provide unpaid support to someone living with a long-term illness, disability or age-related need. This support, which can mean helping someone wash, take medication, move around their home or simply cope, has been valued at £162 billion per year, a 29% increase since 2011 (Centre for Care and Carers UK, 2023). That figure is roughly comparable to the entire annual NHS budget. Yet unpaid-care generates no wages, no employment records and no market transactions. It is largely invisible to conventional economic measurement,

and the only data source that captures where it falls heaviest at neighbourhood scale is the census.

Between 2011 and 2021, the national numbers told a contradictory story. The overall proportion of residents providing any unpaid-care fell from 11.4% to 9.0%, yet the proportion providing 20 to 49 hours per week rose from 1.5% to 1.9%, and those providing 50 or more hours rose slightly from 2.7% to 2.8% (ONS, 2023). Fewer people appear to be caring overall, but those who are caring are doing considerably more of it. Brimblecombe et al. (2018) show that this kind of intensive caring tends to concentrate among those least able to combine it with paid work, which makes understanding where that burden falls a matter of direct relevance to health, employment and social care policy. This project was built to map that divergence and find the places where it has appeared most sharply.

A national statistic alone cannot answer the questions that matter most to the people who plan and commission care services. How did care rates change across Local Authorities between 2011 and 2021? What neighbourhood-level variation lies hidden beneath those averages? And which areas show care intensity that is unexpectedly high relative to their demographic and economic context, rather than simply high in absolute terms? Answering these questions means moving between scales and types of analysis in a way that a single map or statistical bulletin cannot support. Visual analytics is designed to do exactly that. Keim et al. (2010) describe it as the combination of automated computational analysis with interactive visualisation, allowing humans to detect patterns at a scale neither method could reach alone. The dashboards follow Shneiderman’s (1996) principle of overview first, then zoom and filter, then details on demand. Users start with a national picture of Local Authority change, select an area they want to inspect and work progressively deeper into neighbourhood patterns, similarity between areas and model-based comparisons. Milligan and Wiles (2010) argue that care is itself inherently geographical, shaped by the places in which it is given and received, which makes this kind of spatial analysis a natural fit for the subject.

In this project, unpaid-care refers to unpaid help or support provided to someone because of a long-term physical or mental health condition, illness or age-related need, as

recorded in the census. It excludes care provided through paid employment and does not cover general childcare or voluntary work unless directly linked to illness, disability or ageing. Two geographic scales are used throughout. Local Authorities provide the unit for 2011 to 2021 comparison, since they are familiar planning units and large enough for change over time to be meaningful. Middle Layer Super Output Areas, referred to throughout as MSOAs, provide the neighbourhood layer for 2021 and reveal the variation that Local Authority averages can conceal. A full MSOA-level comparison between 2011 and 2021 was avoided because MSOAs are small enough that differences in census question wording or population denominators would make change look larger or more precise than it really is. Economic activity is analysed alongside unpaid-care as a descriptive area-level variable rather than a cause. Because the data is aggregated by area, the analysis cannot identify the circumstances of individual residents.

The intended users are social care commissioners, public health analysts, Local Authority planning teams and academic researchers. Their needs differ in important ways. A commissioner needs to identify neighbourhoods where intensive care burden is unusually high relative to the area’s demographic profile rather than simply high in absolute terms, because high intensive care rates in an older and sicker area are broadly expected, whereas equally high rates in a younger and healthier area are more surprising and may need closer attention. An analyst needs to compare Local Authority trends across a decade and explore how neighbourhood patterns of care relate to health and demographic variables. A researcher may want to examine the neighbourhood clusters and model results in more detail. Existing tools such as ONS Census Maps, Nomis downloads and published statistical bulletins handle single-variable, single-scale questions well but do not allow users to move freely between temporal change, neighbourhood distributions, similarity-based area groupings and model-derived comparisons within one system. The dashboard is designed to fill that gap.

Two research questions structure the project. The first asks how unpaid-care prevalence, unpaid-care intensity and area-level economic activity changed across Local Authorities in England and Wales between 2011 and 2021, and where the contrast between falling overall care and rising intensive care was sharpest. The second asks what neighbourhood-level variation is concealed beneath Local Authority averages in 2021, and which MSOAs show unusually high or low intensive care rates relative to their demographic and economic context. To answer the second question, three dimensionality reduction methods, Principal Component Analysis, t-SNE and UMAP, are applied to MSOA-level data to reveal neighbourhood types and cluster structure. A Bayesian Ridge regression model then estimates expected intensive care rates from area-level context and identifies neighbourhoods where observed care differs substantially from what the model would predict.

Both are described in full in the Data Preparation and Abstraction section.

II. DATA PREPARATION AND ABSTRACTION

The two research questions require different geographic scales and different forms of derived data. The 2011 to 2021 comparison needs a harmonised Local Authority layer with stable boundaries and comparable rates. The neighbourhood analysis needs a 2021 MSOA layer with fields for similarity, cluster membership and expected-versus-observed care intensity. The data preparation pipeline therefore turned raw Census topic counts into analytical layers designed for the dashboard, rather than simply cleaning the files.

Percentage-point changes allow users to compare Local Authorities and spot unusual change. MSOA summary fields show how much neighbourhood variation sits inside each Local Authority. Projection coordinates allow users to explore similarity between neighbourhood profiles. Bayesian residuals identify MSOAs where intensive care is higher or lower than expected for their area context. Figure 1 summarises this workflow, while Table II shows how the prepared layers support the main dashboard tasks.

A. Data sources, audit and variable selection

The project draws on 15 Census CSV extracts spanning four analytical themes, covering demographic structure, ethnic composition, unpaid-care and economic activity. Age and sex were initially considered as separate analytical themes, but were retained as demographic context variables to avoid scope creep and keep the analysis focused on unpaid-care and economic activity. The themes were selected to link care intensity to the demographic and labour-market context most plausibly associated with caring patterns.

A review of the 15 Census files found no missing numeric values, duplicate geography codes or negative values. Unpaid-care was measured in three ways: any unpaid-care, 20 or more hours per week and 50 or more hours per week. Any-care rates are used for descriptive mapping, while 20+ hours care is used as the main modelling outcome because it better captures intensive caring.

Ethnicity was grouped into five broad categories: White, Mixed, Asian, Black and Other ethnic group. A Simpson-style ethnic diversity index was also calculated:

$$D = 1 - \sum p_i^2, \quad (1)$$

where p_i is each group’s share of the local population. The index is used only to summarise how mixed an area is across broad ethnic groups. It is not a measure of identity, integration or deprivation.

Economic activity was summarised using activity, employment, unemployment and inactivity rates. For 2021, economic inactivity was also split into retirement, study, looking after home or family, and long-term sickness or disability (ONS, 2022).

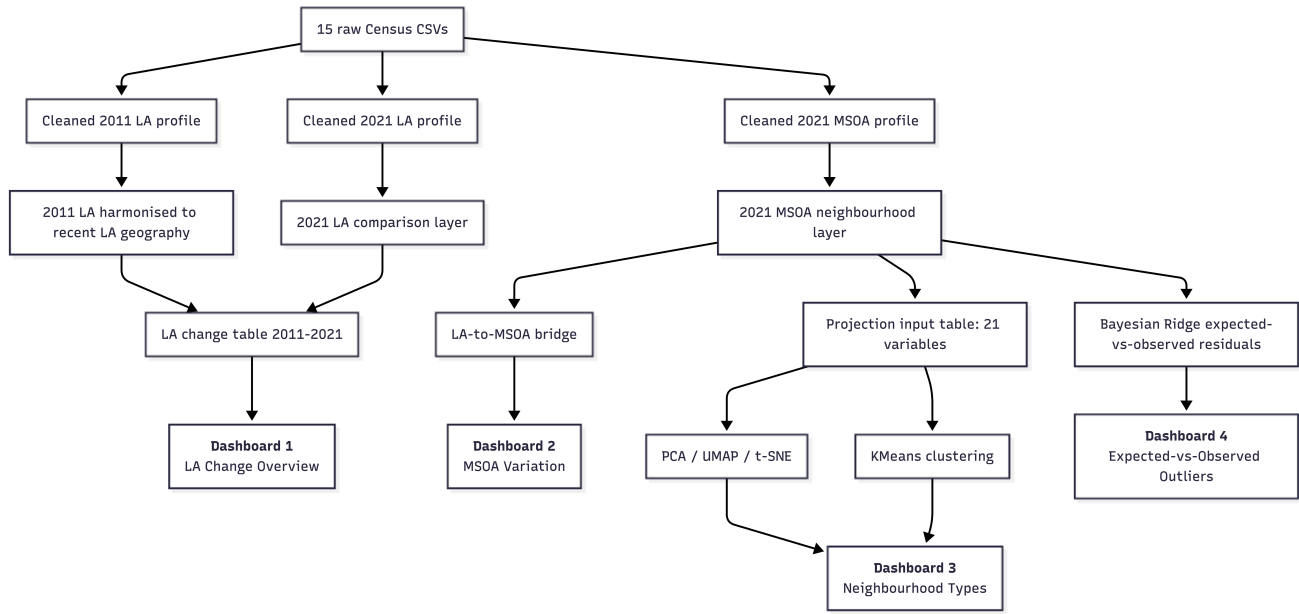


Fig. 1. Multi-scale workflow from raw Census tables to Tableau outputs. Local Authority data supports 2011–2021 change analysis, while 2021 MSOA data supports neighbourhood variation, projections, clusters and Bayesian residuals.

Applying one denominator across all topics with different eligible populations would have produced biased rates. Therefore, all percentages were calculated using topic-specific denominators rather than a single universal population base. Age percentages used the age-table population, ethnicity percentages used the ethnicity-table population, and economic activity percentages used residents aged 16 and over. Unpaid-care percentages used the relevant care-table population, except in the harmonised Local Authority change layer described below, where the 2021 rate was re-expressed using the all-age population for comparability with 2011.

B. Harmonisation, denominator treatment and the LA to MSOA bridge

Two structural issues had to be fixed before comparing 2011 with 2021. The first was that Local Authority boundaries changed between the two censuses. The 2011 data had 348 Local Authorities, while the 2021 data had 331. This means that some areas were merged or reorganised, so a direct comparison could wrongly treat boundary change as socio-economic change. To make the comparison fairer, the 2011 data was converted onto 2021 Local Authority boundaries using the ONS lookup. Where several 2011 authorities matched one 2021 authority, their raw counts were added together first, and percentages were recalculated afterwards. This matters because averaging percentages from differently sized areas would give distorted rates. After this step, both the 2011 and 2021 Local Authority layers had 331 rows, so change over time was calculated using a consistent 2021 boundary structure.

The second issue was definitional. The unpaid-care denominator changed between censuses. In 2011, unpaid-care rates used all usual residents, while in 2021 they used only residents aged five and over. To make the Local Authority change layer more comparable, the 2021 unpaid-care counts were re-expressed using the all-age 2021 population before calculating rates. This makes the two years more comparable, although not perfectly comparable, because the 2021 care question/survey still excluded children under five. For the 2021 MSOA layer, the official aged-five-and-over denominator was kept because this layer is used only for neighbourhood analysis, not for 2011–2021 change.

A second lookup connected the 7,264 MSOAs to their 331 parent Local Authorities, allowing the MSOA layer to be summarised back to Local Authority level. The bridge table then summarised, for each Local Authority, the number of MSOAs and the mean, median, minimum, maximum and interquartile range of MSOA 20+ hours care rates. It also retained equivalent any-care summaries for descriptive comparison and selected contextual averages including 65+ share, economic inactivity, looking after home or family, and long-term sick or disabled. The interquartile range is the central field for the hidden-variation argument because it captures the spread of MSOA care rates inside each Local Authority. It is preferred to a standard deviation because it is less sensitive to extreme MSOA values. MSOAs in the top quarter of national MSOA 20+ hours care rates were flagged as high-care for the LA-to-MSOA bridge. This high-care flag is descriptive and is kept separate from the Bayesian residual measure.

C. Projection and clustering

The MSOA neighbourhood profile uses 21 variables covering age structure, sex composition, unpaid-care, economic activity and broad ethnic composition. The aim was to create an interpretable socio-care profile for each MSOA, rather than a minimal predictive feature set. Some related variables were kept because they are meaningful for dashboard interpretation, even though they partly overlap. For example, economic activity and inactivity are complements, age 65+ overlaps with retirement, and 20+ hours care overlaps with 50+ hours care. This means the projections partly pick up the same demographic signal more than once, so they are treated as exploratory summaries rather than precise measures of similarity. A less redundant feature set would be a useful sensitivity check in future work. All variables were then z-standardised so that variables with larger numerical ranges did not dominate.

Three projection methods were used because they show different aspects of the MSOA profile. PCA was used as an interpretable linear baseline because its components can be linked back to the original variables. UMAP and t-SNE were used as nonlinear projections to show neighbourhood similarity and possible cluster structure, with t-SNE placing the strongest emphasis on local neighbourhoods. The settings and quality metrics are reported in Appendix A. PCA explained 64.4% of the variance across two components. In the plotted orientation, PC1 mainly separated older, less ethnically mixed areas from younger urban areas while PC2 mainly separated areas with higher intensive care, economic inactivity and long-term sickness from areas with higher employment.

Projection quality was checked using two local-neighbourhood measures (trustworthiness and k-nearest-neighbour overlap) calculated at $k = 10, 30$ and 50 . t-SNE preserved local neighbourhoods best, UMAP was close behind, and PCA had the weakest local-neighbourhood preservation but was easier to interpret. KMeans clustering was then applied to the same standardised MSOA profile. Several values of k were compared using silhouette score, Calinski-Harabasz, Davies-Bouldin score and cluster-size balance. The final choice of $k = 5$ was based on a balance between clear profiles, reasonable cluster sizes and stability, rather than the best score on one metric alone.

The silhouette score of 0.24 suggests weak to moderate separation, so the clusters are treated as descriptive area types rather than fixed or definitive categories. The assignments were stable across five random seeds, with adjusted Rand indices between 0.97 and 1.00. Cluster labels and sizes are shown in Table I, with full profiles in Appendix B.

Because the five socio-care clusters include unpaid-care variables, they are interpreted descriptively rather than as evidence that context alone explains care rates. Non-circular context comparison is handled through the Bayesian residual model, which excludes unpaid-care pre-

TABLE I
KMEANS CLUSTER LABELS AND SIZES AT $k = 5$. FULL MEAN PROFILES
IN APPENDIX B.

Cluster	Label	MSOAs
1	Diverse younger urban areas	1,198
2	Older retirement-heavy areas	2,450
3	High-care health-limited areas	1,306
4	Mixed working-age areas	2,139
5	Student-heavy low-care areas	171

dictors.

D. Bayesian residual modelling

The final analytical layer estimates an expected 20+ hours unpaid-care rate for each MSOA, then compares it with the observed rate. The aim is to separate areas that are simply high-care in absolute terms from areas that are high-care relative to their demographic structure. The residual field is exploratory and not causal. It identifies neighbourhoods worth closer inspection, not service failure or unmet need.

Bayesian Ridge regression was selected because it produces a clear expected-versus-observed measure and an uncertainty estimate for each MSOA. Regularisation was useful because several predictors are related to each other. Gradient Boosting was used as a nonlinear benchmark and predicted slightly better than Bayesian Ridge, with a test R^2 of 0.85 compared with 0.80 for the 20+ hours care target. However, it was harder to explain in the dashboard, so Bayesian Ridge was used for the residual layer. Full model performance is reported in Appendix C.

Unpaid-care variables were excluded from the predictor list before model fitting. The model predicted 20+ hours unpaid-care from context variables only: age structure, sex mix, economic inactivity components and ethnic composition. This avoided circularity, so the residual compared observed care against area context rather than against other care measures. Two targets were modelled: 20+ hours unpaid-care and any-care rates. The 20+ hours unpaid-care model was used for the dashboard because it captures intensive caring more directly. A 75/25 train-test split was used for performance assessment, then the final model was refitted on all 7,264 MSOAs to generate the Tableau fields. The residuals shown in Tableau are therefore fitted model differences, not forecasts for new areas.

The residual is measured in percentage points:

$$r = y_{\text{obs}} - y_{\text{exp}},$$

where y_{obs} is the observed 20+ hours care rate and y_{exp} is the model-expected rate.

A positive residual means an MSOA has more intensive unpaid-care than expected for its area context. For each MSOA, the Tableau data includes the expected value, an approximate 95% prediction range, the residual, a residual z-score and a top-or-bottom 5% residual flag. This flag

identifies the most unusual MSOAs for closer inspection. It is not a formal significance test. Out of 7,264 MSOAs, 364 were flagged as higher than expected and 364 as lower than expected.

For the dashboard colours, a simpler rule was used than the top-or-bottom 5% flag. MSOAs more than 1 percentage point above the model-expected rate are shown as higher than expected. MSOAs more than 1 percentage point below it are shown as lower than expected. MSOAs within 1 percentage point either way are treated as close to expected.

E. Interpretation limits

Three limits shape how all findings in this project should be read.

The analysis is entirely area-level, so high unpaid-care and high economic inactivity appearing in the same neighbourhood does not mean the same residents are both carers and economically inactive. In addition, projection coordinates and cluster labels are simplified summaries of complex profiles rather than natural or fixed categories, and the Bayesian residuals show where care departs from model expectation without explaining why. Unmeasured factors such as formal service provision, housing conditions and local care infrastructure may account for much of the residual variation.

Lastly, there is also a minor modelling constraint worth noting. Because the Bayesian Ridge model is linear and the outcome is a bounded percentage, a small number of lower prediction bounds fall below zero. These are not meaningful care rates and affect only the displayed uncertainty range, not the residual ranking or the observed-versus-expected comparison itself. A future version could use a bounded percentage model or a spatial model to handle this more naturally.

III. TASK DEFINITION

The four dashboards serve users who come to unpaid-care data with different professional needs. Social-care commissioners need to identify neighbourhoods where intensive care is unusually high relative to local context rather than simply high in absolute terms, because high rates in an area with an older population and higher long-term sickness or disability are more expected than the same rates in a younger area with fewer health limitations. Public-health analysts need to track how Local Authority care rates changed across a decade and understand how neighbourhood distributions relate to health and demographic variables. Researchers may want to explore cluster structures and unusual model results in more depth. Despite these different starting points, these users all need to move between geographic scales without treating neighbourhood patterns as evidence about individual carers.

The two research questions map directly onto this structure. RQ1 is addressed through Task 1, which examines Local Authority change between 2011 and 2021. RQ2 is

addressed through Tasks 2 to 4, which progressively uncover the neighbourhood variation that Local Authority averages conceal and identify which areas deviate from what their demographic and economic context would predict.

Munzner’s (2014) framework shapes how each task is defined because it separates the domain question from the analytic action needed to answer it. A question like “where is unpaid-care high?” is not a single task. It might mean finding Local Authorities with large change, looking inside a selected authority for neighbourhood variation, exploring similarity between MSOAs across many variables, or finding areas where observed care diverges from model expectation. Each of those requires a different starting point and different forms of visual evidence. Figure 2 summarises the four-stage task flow and shows how each dashboard passes context to the next.

The workbook follows Shneiderman’s (1996) overview-first principle but extends it across four dashboards rather than a single screen. Dashboard 1 provides the national view of Local Authority change. Dashboard 2 filters into a selected authority and reveals the MSOA distribution beneath its average. Dashboards 3 and 4 both support deeper MSOA-level analysis, but they were kept separate because combining projection clusters and model residuals on one screen created visual clutter and asked users to switch between two cognitively distinct tasks at once. Dashboard 3 shows which MSOAs have similar 2021 socio-care profiles, using projections and cluster summaries. Dashboard 4 focuses on model residuals, helping users find MSOAs where observed intensive care is higher or lower than expected. Keeping them apart gave each view enough room to be read clearly without competing for attention.

The tasks also differ in what the user already knows when they start. Munzner (2014) distinguishes search types by whether the target and its location are known in advance. Task 1 is an Explore task because neither the Local Authority nor the pattern of change is known beforehand. Task 2 is a Browse task because the Local Authority is selected but its internal MSOA distribution is not yet known. Task 3 is another Explore task because users are looking for patterns among MSOAs with similar 2021 socio-care profiles, rather than looking for places that are geographically close. Task 4 is a Locate and Identify task because the user knows the type of target they want, a higher- or lower-than-expected residual, but does not yet know which MSOAs qualify. Table II summarises all four tasks using Munzner’s vocabulary alongside the visual evidence designed to support each one.

For Task 1, success means identifying a Local Authority with a large absolute 2011–2021 percentage-point change and backing that choice with evidence from the ranked bar chart, the change map or the scatterplot. For Task 2, success means judging whether the selected Local Authority is internally uniform or varied, drawing on the interquartile range, the MSOA care map, the distribution shape or the

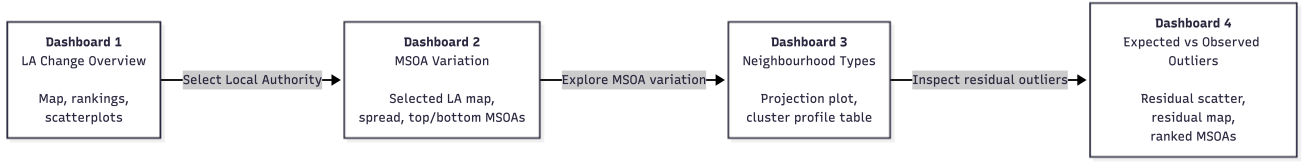


Fig. 2. Four-dashboard workflow from Local Authority change overview to MSOA variation, neighbourhood clusters and Bayesian residual outliers. Selections pass context from one stage to the next.

TABLE II
DASHBOARD TASKS UNDER MUNZNER’S TASK ABSTRACTION.

Task	Munzner abstraction	Visual evidence
Compare Local Authority change	Explore and compare trends and outliers	Change map, ranked bar, scatterplots
Inspect MSOA variation within selected LA	Browse known subset and summarise distribution	MSOA map, interquartile range, distribution, high/low tables
Interpret neighbourhood clusters	Explore similarities and summarise groups	Projection plot, cluster colours, profile table
Find higher- or lower-than-expected MSOAs	Locate and identify outliers	Residual map, observed-vs-expected scatter, outlier table

highest and lowest MSOAs by intensive care tables. For Task 3, success means identifying a cluster and explaining its label using the profile table, while reading projection proximity as similarity across the selected 2021 socio-care variables rather than geographic closeness. For Task 4, success means finding a valid residual outlier, reporting the observed and model-expected values together and framing the result as a contextual deviation rather than a causal finding.

Each task also carries an interpretation risk and the dashboard design aims to reduce these rather than ignore them. Local Authority change maps can hide neighbourhood variation, so the workbook links the overview to within-authority MSOA distributions. MSOA distributions can be misread as evidence about individual carers, so labels and tooltips use area-level wording throughout. Projection plots can be mistaken for geographic maps, so the axes are labelled as projection dimensions and the cluster profile table explains what each group represents. Residual maps can be misread as evidence of service failure or unmet need, so the residual layer is framed as exploratory and the prediction range is shown alongside the observed and expected values.

IV. VISUALISATION JUSTIFICATION

Every design choice in the workbook is made to support a specific task and guard against a specific misreading. Munzner’s (2014) framework guides the justification throughout, matching each encoding to the data type, the analytic task and the perceptual limits of the reader. Each dashboard anchors a different stage of the analysis. Dashboard 1 ensures the analysis begins from a national picture rather than an isolated example. Dashboard 2 prevents a Local Authority average from being read as though the whole authority behaves the same way. Dashboard 3 shows neighbourhood similarity across many variables at once rather than keeping

the analysis purely geographic. Dashboard 4 separates areas that are simply high-care from those where care is high relative to what their local context would predict. Placing all four on a single screen would require users to switch between cognitively distinct tasks at once, and that is where misreading tends to begin.

A. Perceptual and cognitive principles

The workbook separates broad pattern recognition from precise comparison throughout. Choropleth maps show where patterns fall geographically but are always paired with ranked bars, scatterplots or tables when users need to compare values or look up exact figures. Position on a shared scale and bar length are more accurately judged than differences in colour saturation, so colour is reserved for spatial overview while position-based charts handle precise comparison.

Visual grouping is used as a starting point rather than a conclusion. Colour separates cluster and residual groups and screen-space proximity helps users see local structure in the projection views. These fast impressions are useful for an initial reading but can encourage over-interpretation, so every visual grouping is backed by explanatory labels, profile tables and tooltips. The projection view states its reading rule directly at the left of the screen because it is the view most prone to misreading. Points that sit close together share similar area profiles rather than being geographically close to one another, and making that explicit is what keeps the view useful.

Colour meaning is held consistent across the whole workbook. Sequential blue encodes positive ordered rates, a diverging orange to blue palette encodes signed values around a meaningful zero such as change scores and residuals, and a separate set of categorical colours encodes the five neighbourhood clusters. A reader does not have to relearn what a colour means when moving between

dashboards. On Dashboard 4, orange marks MSOAs more than one percentage point above model expectation, blue marks those more than one percentage point below, and grey covers the near-expected majority. Cluster identity is also encoded redundantly through colour, legend label and table row so that groups can be distinguished without relying on colour alone, which matters for accessibility.

Reference lines and text hierarchy reduce the mental effort of reading each view. Solid diagonal lines mark equality or no-change benchmarks, solid zero lines mark increase and decrease boundaries, and dashed lines are reserved for secondary trend relationships. Dashed rather than dotted lines were chosen because dotted marks compete visually with scatterplot points. Mean benchmark lines are added wherever a distribution needs context so users can judge whether a selected area sits above, below or near a wider average rather than reading the spread in isolation.

Dashboard titles are written as questions to state the task directly. Within subtitles, lighter supporting wording is combined with bold key variables, years and dashboard names, so terms such as **20+ unpaid-care rate, Dashboard 1** and **2011–2021** are noticed first. Legends explain colour meaning and tooltips supply exact values on hover. The aim throughout is for every piece of text to guide interpretation rather than add visual noise.

B. Dashboard 1, Local Authority change overview

Dashboard 1 addresses the first research question and supports the Explore task. Three view types work together. The shaded map gives the national overview and includes a selector allowing users to switch between the 2011 rate, the 2021 rate and the 2011 to 2021 change. The two rate views share the same sequential colour scale for direct comparability, while the change view uses a zero-centred diverging scale so that increases and decreases read clearly in opposite directions. The ranked bar chart then provides the precise comparison that colour on a map cannot, because bar length is decoded more accurately than colour saturation. Together the two views let users move from “where is care concentrated?” to “which areas changed most?” without leaving the screen.

The care-measure selector switches between 20+ hours intensive care and any unpaid-care. As intensive care rose between 2011 and 2021 while overall care prevalence fell, a single measure would hide that divergence, so the selector allows prevalence and intensity to be examined separately without conflating them in one encoding.

Two scatterplots support relational comparison. In the 2011-versus-2021 view, the solid diagonal shows which Local Authorities changed relative to their starting point, while the dashed trend line shows the overall direction across all authorities. In the care-change versus economic-inactivity-change view, solid zero lines divide the chart into four quadrants so users can easily find areas where both measures moved in the same direction. A second

ranked bar shows the interquartile range of MSA 20+ hours care rates within each Local Authority, introducing the neighbourhood variation argument that Dashboard 2 develops.

A population-weighted cartogram would reduce the visual weight given to large rural areas relative to their population size, but it would also make the geography harder to recognise. The familiar geographic map was therefore kept and paired with ranked bars and scatterplots to enable accurate comparison alongside a spatial overview.

C. Dashboard 2, MSA variation within the selected Local Authority

Dashboard 2 shifts the unit of analysis from the Local Authority as a whole to the neighbourhoods inside it. The question changes from whether an authority is high or low to how much variation sits beneath its average. The MSA map and the distribution plot complement each other because each covers the weakness of the other.

The map shows where high and low care MSAs sit geographically, while the distribution plot conveys the spread of rates more precisely than colour differences on a map can manage. Benchmark lines for the selected Local Authority mean and the England and Wales MSA mean give the spread an interpretive frame, allowing users to judge whether the selected authority is typical or unusual rather than reading the distribution in isolation.

The interquartile range is used as the primary measure of spread because it is less affected by extreme MSA values than a standard deviation. The highest and lowest MSAs by intensive care tables provide exact neighbourhood names and rates for direct lookup. The summary panel on the left preserves the selected authority context as users work deeper into the detail. Together these views make the central argument of this dashboard visible, which is that a Local Authority average can hide a wide range of neighbourhood experience underneath it.

D. Dashboard 3, neighbourhood types and projections

Dashboard 3 supports the task of comparing MSAs across many variables at once. Each point in the projection represents one MSA, positioned by its similarity across the 21 standardised socio-care variables used in the analysis. The reading rule is stated at the left of the dashboard because projection space is the view most likely to be misread. Points that sit close together share similar area profiles and are not necessarily geographically close, and being clear about that distinction is what allows the view to be interpreted correctly.

Three projection methods are offered because each reveals a different aspect of the similarity structure. PCA serves as the interpretable linear baseline since its components can be linked back to the original variables, making it easier to understand what the axes represent. UMAP and t-SNE preserve local neighbourhood structure better than PCA but both can distort distances between far apart

clusters (van der Maaten and Hinton, 2008; McInnes et al., 2018), and both respond to their algorithm settings in ways that can produce apparent gaps between groups which reflect the method rather than the data itself. Showing all three therefore gives users a more robust picture than any single projection alone would provide.

The cluster profile table beneath the projection is what makes the view interpretable in practice. A coloured scatterplot can look convincing without telling users what the colours mean in substantive terms, so the table translates each numbered cluster label into the average MSOA characteristics for that group, covering care rates, age structure, economic inactivity, student share and ethnic mix. The wording makes clear that these are cluster averages rather than percentage contributions to care, which was a source of confusion in the evaluation. The projection selector and cluster filter let users compare the three methods and show or hide specific neighbourhood types without changing the underlying cluster definitions.

E. Dashboard 4, Bayesian expected against observed care

Dashboard 4 separates MSOAs that are high in absolute terms from those that are high relative to what their demographic context would predict. The model-expected 20+ hours care rate for each MSOA comes from the Bayesian Ridge model, which bases its estimate on age structure, sex composition, work-status profile and ethnic mix. The observed-versus-expected scatterplot is the conceptual anchor for the whole dashboard. A single solid diagonal marks the line where observed and expected care are equal, so points above the line carry more intensive care than the model predicts and points below carry less. This makes the residual idea visible through position in the scatterplot rather than something users have to infer from a number alone. Group-specific trend lines are not included because the task is to compare every MSOA against the same model expectation rather than to fit relationships.

The residual map places the scatterplot findings back into geographic context. Each view guards against a different misreading. The scatterplot ensures that residual colours are understood in relation to the model baseline before conclusions are drawn, while the map prevents residuals from being treated as abstract points with no location. Both views use the same orange, grey and blue categories so colour meaning stays consistent. Orange marks MSOAs more than one percentage point above expectation, blue marks those more than one percentage point below, and grey covers the large majority that fall close to expectation.

The audit table remains fixed on the strongest higher-than-expected residual cases regardless of what users are exploring in the map or scatterplot, keeping its purpose clear and stable. The dashboard wording is careful to frame residuals as areas where observed care exceeds or falls short of the model’s expectation rather than as evidence of service failure or unmet need. The audit table gives the observed

value, the expected value, the residual, the 95% uncertainty range and supporting context variables.

F. Layout, interaction and navigation

All four dashboards share the same layout, with a question-style title and task prompt at the top, analytic views in the centre, and controls and the legend on the right. Keeping this structure consistent means users do not have to relearn where to look when moving between dashboards, which reduces the overhead of navigating a multi-view workbook. Navigation buttons follow the same placement pattern throughout, reinforcing the sequence from Local Authority overview through neighbourhood detail, cluster analysis and model residuals.

Interaction is designed to match the specific task of each dashboard rather than being applied uniformly. Selecting a Local Authority on Dashboard 1 passes that choice to Dashboard 2, creating the main overview-to-detail transition. Dashboard 3 uses projection and cluster controls to show or hide neighbourhood types. Dashboard 4 preserves the full national residual picture rather than filtering down to a selection, because the value of the residual map depends on being able to see where the exceptions sit relative to the whole country. Tooltips on every chart supply the area name, the relevant value and supporting context, giving users an extra and precise lookup layer.

V. EVALUATION

A. Framework and design

The evaluation tested whether participants could complete the full four-dashboard workflow, moving from identifying a Local Authority with notable change in 20+ hours unpaid-care, through inspecting neighbourhood variation within it, interpreting a neighbourhood cluster, and finally locating an MSOA where observed care exceeded the model’s expectation.

The approach drew on Munzner’s (2014) four-level model, which meant the evaluation produced stronger evidence in some areas than others. It gave the clearest picture at the encoding and interaction level, testing whether users understood the charts, colours, controls and overall layout. It gave partial evidence at the task and data abstraction level, since several difficulties arose from unclear derived measures rather than from interaction problems alone. Evidence at the domain level was weaker because participants were fellow students rather than professional commissioners or planners. Algorithm quality was assessed separately through projection quality metrics, cluster stability tests and model performance checks reported in Appendices A and C.

Three participants completed the tasks in order using a think-aloud method. The observer recorded task completion, confidence scores (1-5), hesitations and any comments. Timing was planned but not recorded consistently.

TABLE III
PARTICIPANT CONFIDENCE SCORES BY TASK.

	Participant A	Participant B	Participant C	Mean
Task 1 — LA change	4	4	4	4.0
Task 2 — MSOA variation	3	4	3	3.3
Task 3 — Cluster interpretation	1	1	2	1.3
Task 4 — Residual outlier	4	3	4	3.7

B. Results

Confidence followed a clear arc across the four tasks, starting high, dropping sharply at the third dashboard and recovering at the end. This happened consistently across all three participants and tracked closely with whether each dashboard made its comparison rule visible. Table III shows the full confidence scores by participant and task.

Task 1, Local Authority change (3/3 completed).

Participant A used the ranked bar chart to find the Isles of Scilly with a 1.44 percentage-point change, while Participants B and C used the scatterplot and selected Hartlepool. The fact that users reached a valid answer through different routes suggests Dashboard 1 offered enough flexibility to suit different working styles. The main issue was task wording. “Substantial change” was too open, and future testing should specify whether the goal is the largest increase/decrease or the largest absolute change.

Task 2, MSOA variation (3/3 partially completed).

Participants could use the map, ranked bars and boxplot to find high and low care MSOAs, and Participant A found the legend helpful for identifying the extremes. The shared difficulty was judging whether the spread they found was meaningfully large or small. As Participant C put it, “I’m not sure what the spread in MSOAs is being compared against and that there needs to be more information about that.” Participant B eventually called the distribution tightly clustered with confidence 4 but admitted uncertainty. Users could find the values but had no reference point to evaluate what they meant.

Task 3, cluster interpretation (0/3 fully completed). No participant fully completed Task 3, and it produced the lowest confidence scores by a large margin. Participant A could not interpret the clustering view at all. Participant B identified the older-retirement cluster correctly and understood that clusters described area types rather than individuals, but misread the profile table, saying “I think the columns show the percentage contribution of each type of care for that group.” That was a reasonable guess but an incorrect one, and the column headers did nothing to prevent it. The core problem was the projection view itself. Users needed clearer guidance on which variables shaped the layout and what it meant for two points to sit close together. Dashboard 3 also broke the reading habit built across the first two dashboards, where users had always started with a summary before drilling into detail. As Participant A observed, “you always need to spend brain resources just to understand the graph,”

suggesting that re-orientation was consuming attention that should have gone towards analysis.

Task 4, residual outlier detection (3/3 completed).

Task 4 recovered strongly. Although Dashboard 1 had the highest mean confidence score, all three participants described Dashboard 4 as the easiest of the four despite its more complex modelling basis. Participant B found Wigan almost immediately, saying “it’s orange on the legend, and orange means higher than expected.” Participant A suggested adding the observed and model-expected values directly to the residual table so the calculation could be verified at a glance. The dashboard worked because colour, the diagonal reference line and the table all reinforced the same comparison rule, leaving users nothing to infer.

C. Design changes

Because the evaluation was formative, the findings shaped targeted revisions to the workbook rather than serving as a final validation.

Comparison rules were made more explicit throughout. Dashboard 1 now distinguishes solid reference lines from dashed trend lines, and the task wording was tightened to focus on the largest absolute percentage-point change rather than the vague phrase “substantial change.” Dashboard 4 replaced separate residual-group trend lines with a single observed-equals-expected benchmark, removing the impression that different residual groups followed meaningfully different relationships. Dashboard 2 gained selected Local Authority and England and Wales MSOA mean lines, which directly addressed the difficulty participants had in judging whether a neighbourhood spread was large or small without any point of comparison.

Dashboard 3 received substantial revisions because it produced the clearest interpretation failure. Projection-reading guidance was added at the left of the view, the cluster table wording was rewritten to make clear that values are cluster averages rather than percentage contributions to care, and cluster labels were shortened and numbered so the projection, legend and table all use the same identifiers. Dashboard 4 had observed and model-expected values added directly alongside the residual and uncertainty range in the audit table, making the underlying calculation visible and checkable. Navigation, titles, subtitles and tooltips were standardised across all four dashboards so each view has a consistent reading path and hover text covers only the fields visible in that particular view.

D. Interpretation and limitations

The clearest lesson from the evaluation was that usability depended on whether the comparison rule was made visible rather than on how technically complex the underlying method was. Dashboard 4 used the most demanding analysis but was the easiest to use because colour, the reference line and the table all pointed to the same observed-versus-expected idea. Dashboard 3 was the hardest not because dimensionality reduction is inherently confusing but because the rules for reading projection distance and cluster profiles were never stated clearly enough for users to apply without guidance.

The evaluation has real limits. The sample was small and consisted of fellow students rather than professional users, the fixed task order may have introduced learning effects, timing data was incomplete and the observer was also the designer of the workbook. The revised version was not re-tested after the changes, so there is no direct evidence that every problem was solved. What the evaluation does reliably establish is which interpretation difficulties appeared, what was changed in response, and why making comparison rules visible became the central design priority after the sessions.

VI. CONCLUSION

Between 2011 and 2021, the headline story for unpaid-care in England and Wales was a falling one. Fewer people were recorded as carers overall, but beneath that decline intensive caring quietly grew. The Isles of Scilly, Hartlepool and Eastbourne all saw 20 or more hours care rise by over a percentage point, and every one of the twelve highest-change Local Authorities also saw economic inactivity increase over the same decade. The divergence was not random. In the highest-change authorities, rising intensive care coincided with rising economic inactivity.

Local Authority averages, however, obscured as much as they revealed. Newcastle upon Tyne had a 20+ hours care rate of 4.4%, fractionally below the England and Wales mean of 4.6%, yet the spread across its neighbourhoods was 3.63 percentage points, the widest of any Local Authority. Kensington and Chelsea (3.25pp) and Brighton and Hove (2.97pp) showed the same pattern, where the average looked unremarkable but the distribution beneath it did not. More strikingly, the highest intensive care rate of 6.5% appeared not in the oldest retirement-heavy areas, which had by far the largest share of residents aged 65+, but in the health-limited cluster where long-term sickness and disability were more than twice the national average. Planning services around age profiles alone would miss the communities carrying the greatest burden.

A further assumption the data challenged was that more ethnically diverse areas carry heavier informal care burdens, as family-based care cultures are sometimes assumed to fill gaps left by formal provision. The three clusters with the highest ethnic diversity, covering diverse younger urban, mixed working-age and student-heavy areas, recorded the

three lowest intensive care rates. The most diverse cluster reached only 4.2% despite having the highest ethnic mix of any group, while the least diverse cluster had a higher intensive care rate despite a sevenfold gap between them in the ethnic diversity index. This does not explain why care rates are lower in more diverse areas, but it does suggest the assumption deserves scrutiny rather than acceptance.

The modelling layer then separated areas with high care in absolute terms from those where care exceeded what their local profile would predict. East Lindsey was the clearest case, with three of its neighbourhoods already among the oldest and most economically inactive in England and Wales, yet still recording 20+ hours care rates 2.4 to 3.1 percentage points above model expectation. Higher-than-expected residuals appeared more concentrated in parts of Wales and the Midlands, while lower-than-expected cases were scattered without any clear geographic pattern. Because the residual flags isolate the highest and lowest 5%, these cases are best read as exceptional neighbourhoods for closer inspection rather than evidence of a widespread national pattern.

The central visualisation lesson was that analytical complexity and usability are not the same thing. The Bayesian residual dashboard was technically the most demanding component yet was the easiest for participants to use, because its comparison rule was clearly stated and observed care was visibly either above, below or close to model expectation. The projection dashboard was the hardest, not because it was more complex, but because it asked users to read distance and cluster profiles without enough guidance on what those things meant. Hence, making the comparison rule explicit matters more than methodological sophistication.

The workbook remains an exploratory tool rather than a prescriptive guide. All findings describe neighbourhood patterns rather than the circumstances of individual carers. The 2011 to 2021 comparison rests on careful but imperfect boundary and denominator harmonisation, and the cluster labels are descriptive area types rather than fixed categories. The final post-evaluation changes were not re-tested, so they should be read as informed design improvements rather than proven usability gains. The most important next steps are validation with professional users, stronger benchmarks that compare each Local Authority with similar places such as nearby areas, equally deprived areas or authorities with similar demographic profiles, and a residual model that accounts for local service provision, because a community's care burden cannot be fully understood without knowing what formal alternatives exist nearby.

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APPENDIX A

PROJECTION SETTINGS AND QUALITY METRICS

TABLE IV
PROJECTION SETTINGS.

Method	Settings
PCA	2 components.
UMAP	2 components; 30 neighbours; minimum distance 0.1; Euclidean metric; random seed 7.
t-SNE	2 components; perplexity 40; automatic learning rate; PCA initialisation; 1,000 iterations; random seed 7.

PCA explained 43.1% of variance on PC1 and 21.2% on PC2, giving a two-component total of 64.4%. In the plotted orientation, PC1 primarily contrasts older, retired and ethnically less heterogeneous areas against younger and more diverse urban areas. PC2 is more closely associated with intensive care, economic inactivity, long-term sickness or disability and looking after home or family, against employment and economic activity. Trustworthiness and k-nearest-neighbour overlap were calculated at $k = 10, 30$ and 50 . t-SNE produced the strongest local-neighbourhood preservation, UMAP was close behind, and PCA was weakest on local fidelity but strongest as an interpretable linear baseline.

TABLE V
PROJECTION QUALITY METRICS FOR LOCAL-NEIGHBOURHOOD PRESERVATION.

Method	k	Trustworthiness	kNN overlap
PCA	10	0.9066	0.0826
PCA	30	0.9082	0.1624
PCA	50	0.9096	0.2152
UMAP	10	0.9632	0.1894
UMAP	30	0.9611	0.2914
UMAP	50	0.9594	0.3433
t-SNE	10	0.9930	0.3660
t-SNE	30	0.9838	0.4013
t-SNE	50	0.9771	0.4162

APPENDIX B

CLUSTER MEAN PROFILES

TABLE VI
KMEANS CLUSTER MEAN PROFILES AT $k = 5$.

Cl.	Label	MSOAs	Any care	20+ care	65+	Retired	LT sick / dis.	Diversity
1	Diverse younger urban areas	1,198	7.4	4.2	10.5	11.6	4.8	0.604
2	Older retirement-heavy areas	2,450	9.9	4.6	26.5	30.6	3.2	0.087
3	High-care health-limited areas	1,306	10.5	6.5	17.8	21.0	7.6	0.163
4	Mixed working-age areas	2,139	8.1	3.8	16.9	19.8	3.1	0.255
5	Student-heavy low-care areas	171	5.4	2.4	9.2	9.5	3.0	0.431

Notes: care, age, retired and long-term sick or disabled values are percentages. LT sick / dis. = economically inactive due to long-term sickness or disability. Diversity is the Simpson-style ethnic diversity index.

Cluster stability across random seeds 7, 21, 42, 100 and 123 produced adjusted Rand indices between 0.9684 and 1.000 against the main seed, with silhouette scores between 0.239 and 0.240.

APPENDIX C

MODEL COMPARISON AND BAYESIAN RIDGE PREDICTORS

The 75/25 train-test split was used for performance assessment only. Final dashboard fields come from the model refit on all 7,264 MSOAs.

TABLE VII
MODEL COMPARISON ON THE 75/25 TRAIN-TEST SPLIT.

Target	Model	Test MAE (pp)	Test R^2
20+ hours care	Bayesian Ridge	0.476	0.798
20+ hours care	Gradient Boosting	0.399	0.854
Any care	Bayesian Ridge	0.646	0.760
Any care	Gradient Boosting	0.572	0.810

TABLE VIII
BAYESIAN RIDGE PREDICTOR GROUPS.

Predictor group	Variables
Age structure	Percentage aged 65+; youth dependency ratio.
Sex composition	Percentage female.
Economic context	Percentage unemployed; retired; student; looking after home or family; long-term sick or disabled.
Ethnic composition	Percentage Mixed; Asian; Black; Other ethnic group; ethnic diversity index.
Excluded variables	Unpaid-care variables were excluded to avoid predicting care from other care measures.

APPENDIX D FINAL TABLEAU OUTPUT FILES

TABLE IX
FINAL TABLEAU OUTPUT FILES.

Output file	Contents	Dashboard role
LA overview table	Harmonised LA rates, 2011–2021 changes and MSOA bridge fields.	Local Authority change dashboard.
MSOA profile table	2021 MSOA variables, projection coordinates, cluster labels and residual fields.	MSOA maps, projections and model views.
Socio-care cluster summary	$k = 5$ cluster profiles and cluster sizes.	Cluster interpretation.
Model comparison summary	Bayesian Ridge and Gradient Boosting performance metrics.	Model justification.
PCA loading table	PCA variable loadings.	Projection interpretation.

Final checks reported 7,264 MSOA rows, 331 Local Authority rows, five main cluster rows and four model-summary rows, with zero missing values in the final MSOA and Local Authority Tableau files.